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- Ubuntu 18.04
- Jenkins

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This document provides a comprehensive guide to deploying a containerized application using Doc

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2. Prerequisites

Hardware Requirements
- Minimum 2 Ubuntu 18.04 servers (1 manager, 1 worker)
- Sufficient CPU, RAM, and disk space based on application needs

Software Requirements
- Ubuntu 18.04 installed on all nodes
- Root or sudo privileges
- Stable internet connection

Network Configuration
- Nodes should be on the same network or VPN
- Open necessary ports:
  - Docker Swarm: 2377 (cluster management), 7946 (communication), 4789 (overlay network)
  - Jenkins: 8080 (default)
  - Monitoring tools ports as needed

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3. Environment Setup

3.1. Installing Docker on Ubuntu 18.04
```bash
sudo apt update
sudo apt install -y apt-transport-https ca-certificates curl software-properties-common
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
sudo add-apt-repository \
   "deb [arch=amd64] https://download.docker.com/linux/ubuntu \
   $(lsb_release -cs) stable"
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io
```

Deployment Guide for 2019 Stack
Stack Components:
- Docker Swarm
- Ubuntu 18.04
- Jenkins

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